Project Nova

CartCoders
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Unlocking the Half-Trillion Future:

The Problem:

- **350 million Southeast Asians** lack access to formal credit despite having income (source)
- Over 60% of Southeast Asians remain underbanked or unbanked (source)
- **$300 billion funding gap** for MSMEs in Southeast Asia
- Traditional credit scoring **excludes gig economy workers** who drive regional economic growth (source)

The Grab Context:

- **Dominant market position:** 55% market share in Southeast Asia food delivery, leading mobility platform.
- **Key markets:** Indonesia (largest), Singapore (headquarters), Thailand (fastest growth), Philippines (emerging)
- 5M+ active partners with **rich behavioral data spanning years**
- Partners earn consistent income but face **25-40% APR informal lending rates**

THE OPPORTUNITY *Transforms Grab's unique ecosystem data into Southeast Asia's largest alternative credit bureau, unlocking access to formal credit for millions across 4 major markets*

The Solution: Trust as Collateral

NovaScore Algorithm:

- **Performance Trust (40%):** Trip completion rates, customer ratings, consistency scoring
- **Financial Behaviour (35%):** Earning volatility analysis, transaction pattern recognition, cash flow modeling
- **Platform Engagement (25%):** Feature adoption depth, growth trajectory, ecosystem participation

NovaScore v/s Traditional Credit Score

Traditional Credit Score Limitations:

- **Backward-looking**: Based on past loan repayment history (often 2-7 years old)
- **Credit Invisible Bias**: 44% of southeast asians adults have no credit history at all (source)
- **Static Updates**: Monthly/quarterly reporting cycles
- **Limited Data**: Only formal banking transaction excludes gig economy income
- **One-criteria-fits-all**: Same criteria for salaried employees and gig workers

NovaScore Innovation:

- **Forward-looking**: Predicts future credit worthiness using real-time behavioural data
- **Inclusive by design**: Works for credit-invisible gig workers with consistent income
- **Real-time Updates**: Score adjusts with every trip, delivery, customer interaction
- **Rich behavioural data**: Score adjusts with every trip, delivery, customer interaction
- **Context-aware**: Understands gig economy income volatility and seasonal patterns

Three-Tier Product Statergy:

Driver-Partner Vehicle Financing:

- **Motorcycle Loans (Southeast Asia):** $500-3,000 for delivery riders, 12-24 months terms
- **Car Loans (Singapore/Thailand)**: $5000,-25,000 for ride-hailing, 24-48 months terms
- **Unique Value:** Real-time earnings data enables dynamic payment scheduling

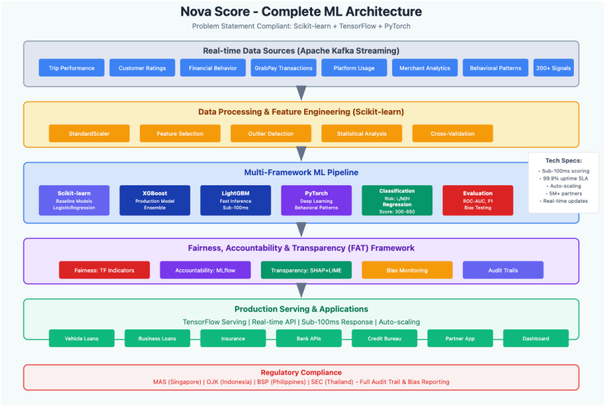
Merchant-Partner Business Expansion:

- **Micro-Loans (Food/Retial):** $200-5,000 for inventory, equipment, marketing
- **Growth Capital:** $5000,-50,000 for second locations, franchise opportunities
- **Integration:** Direct integration with GrabFoods/GrabMart sales analytics

Personalised Financial Ecosystem:

- **Insurance Products:** Usage-based auto insurance, health coverage via GrabCare
- **Savings & Investments:** Goal-based savings linked to GrabRewards, micro-investing
- **B2B Services:** Fleet management loans, merchant payment solutions

NovaScore Tech-Architecture



Machine Learning Stack:

- **Data Processing:** Scikit-learn preprocessing, feature scaling, statistical analysis
- **Baseline Models:** Scikit-learn logistic regression and random forest for comparison
- **Primary Models:** XGBoost ensemble with LightGBM for production inference
- **Deep Learning:** PyTorch neural networks for complex behavioural pattern recognition
- **Dual Output:** Classification (Risk: Low/Medium/High) + Regression (Score: 300-850)
- **Feature Engineering:** 200+ behavioural signals processed via Apache Kafka streams
- **Model Serving:** TensorFlow Serving with sub-100ms response times

Fairness, Accountability & Transparency (FAT) Framework:

- **Fairness:** TensorFlow Fairness Indicators, adversarial debiasing, statistical parity testing
- **Accountability:** MLflow model versioning, comprehensive audit trails, regulatory compliance logging
- **Transparency:** SHAP global explanations + LIME local interpretability for every decision

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Score Comparison Examples:

Traditional Credit Score Card:

- **Partner:** Raj Singh (Delhi)
- **CIBIL Score:** 550 → High Risk ❌
- **Status:** Rejected (insufficient credit history)
- **Available Products:** None

NovaScore Card - Growth Journey:

- **Partner:** Ahmed(Jakarta, 6-months progression)
- **Month 1:** NovaScore 420 → No eligibility
- **Month 3:** NovaScore 520 → Micro-loan Rp. 5,000,000 @16% APR
- **Month 6:** NovaScore 680 → Vehicle upgrade loan Rp. 25,000,000 @12% APR
- **Growth drivers:** Improved Ratings (3.9 → 4.7), consistency +85%, peak-hour availability.
- **Financial Impact:** 40% earnings increase through better vehicle + loan access

NovaScore Card - Established Partners:

- **Partner:** Raj Singh (Delhi, 2 years on platform)
- **NovaScore:** 720 → Low Risk ✅
- **Journey:** Started at 450 → Now top 15% of partners.
- **Key drivers:** Earnings consistency, Avg. ratings 4.8/5, 5% cancellation rates
- **Current offers:** Pre-approved vehicle loan ₹2,50,000 @ 12% APR
- **Premium Access:** Insurance discounts, priority support, business expansion credit line

Why Now?

- Regulatory push for financial inclusion across ASEAN
- Growing acceptance of alternative credit data
- Post-Pandemic focus on gig economy worker rights
- Grab's expanding financial services ecosystem

Fairness-First Technical Approach:

- **Adversarial Debiasing:** Neural networks trained to remove demographic correlations
- **Equitable Adjustment Engine:** Post-Processing algorithms ensuring equal opportunity across groups
- **Real-time Bias Monitoring:** Continuous A/B testing with fairness metrics dashboard
- **Explainable AI:** SHAP (Shapely Additive Explanations) Values and LIME (Local Interpretable Model-agnostic Explanations) for every scoring decision

Business Model & Revenue Streams:

- **Lending-as-a-Service:** 1-3% organisation fees from partner banks/NBFCs
- **Direct Grab Financial Products:** 8-18% APR (v/s 25-40% informal market)
- **NovaScore API Licensing:** $0.10 per score to external platform
- **Risk Assessment Services:** Insurance premium, optimisation, fraud detection

Market Specific Rollout:

- **Indonesia:** Focus on motorcycles loans(2M+ drivers), partners with local banks
- **Singapore:** Car loans and merchant expansion, regulatory sandbox participation
- **Thailand:** Rapid deployment leveraging strong growth momentum
- **Philippines:** Micro-lending focus, partnership with rural banks

Projected Impact & Metrics (estimated):

- **Year 1:** 100K partners scored, $50M loans facilitated, 15% default rate(v/s 25% traditional)
- **Year 3:** 2M partners, $500M lending volume, $25M revenue, 8 country expansion
- **Social Impact:** 40% increase in partner income, 300k new financial products users

(Projections based on validated market sizing, weighted average loan calculations, and benchmarked against existing Grab Financial performance)

Global Competitive Landscape:

- **vs. Uber(Global):** Only offers basic debit card and third-party lending partnerships: no proprietary credit scoring (source)
- **vs. GoTO Financial (Indonesia only):** NovaScore spans 8 countries, 3x behavioural data richness
- **vs. Traditional Banks:** 95% faster approval(minutes vs weeks), 50% lower default rates (source)
- **vs. Existing Credit Bureaus:** Real-time behavioural data vs historical financial records only (source)

Technical Competitive Moats:

- **Global First-Mover Advantage:** Uber drivers still require traditional 600+ credit scores for loans
- **Data Advantage:** Only Performs with cross-borders gig economy behavioural data
- **Real-time Architecture:** Sub-100ms scoring vs traditional bureau's days/weeks
- **Fairness-by-Design:** Patent-pending bias mitigation algorithms
- **Regulatory Compliance:** Built for MAS, OJK, BSP, financial inclusion, requirements